PORTFOLIO

Faishal Yusuf Baqir · Software Developer Project Officer · Flutter Mobile Developer · github.com/faishalyb

1. DO! IAM — Plantation Information Management System

PT Lentera Kreasi Teknologi (First Resources Ltd) · Offline-first Flutter app for 3 palm oil groups (FR, FAP, KAS) Nov 2025 – Present

A new mobile system, built from scratch, that harvesters, supervisors, field assistants and verifiers use every day in estates with little or no signal. It replaces paper for attendance, harvest, transport, inspection and nursery work, stores everything in SQLite on the device and syncs when a connection is available. Built by a team of 3 developers; I wrote about 55% of the commits across both phases.

	Phase 1	Phase 2
Period	Nov 2025 – Oct 2026 (v3.0.11)	Jun 2026 – Present (v1.0.0 Phase 2)
Field data	**NFC**: harvest and transport data written to and read from NFC tags	**Bluetooth thermal printer**: printed harvest tickets, reprint and recap, SPB
Face recognition	FaceNet-512 (~47 MB) + MiniFASNet liveness	**ArcFace-based MobileFaceNet (w600k_mbf, 6.9 MB)** + face alignment + MiniFASNet liveness + Hikvision face sync
Modules	Attendance, harvest (panen) and harvest estimation, SPB, mill, BKM, job assignment, employee transfer, 6 inspections (panen, pupuk, tunas, gawangan, 2 spraying)	Taksasi panen, timbang peron, sortasi, nursery (seed receipt, selection, transplanting), vehicle monitoring, LP3, audit trail, 6 inspections (APD, bibitan, mutu buah, panen, perawatan, tunas progresif) with verification and bulk delete
Platform	Offline login, master sync and transaction upload, GPS tracking and geofence, backup and restore, auto-destroy of local data, in-app APK update check	

Face recognition upgrade (Phase 2)

Field users reported false "face already registered" warnings, wrong names at check-in and other people passing face login. I benchmarked three models, chose an ArcFace-based MobileFaceNet, converted it to TFLite, added face alignment to the ArcFace 112 px template and recalibrated thresholds.

Model	Size	TAR @ FAR 1e-3	TAR @ FAR 1e-4	TAR @ FAR 1e-5
FaceNet-512 (Phase 1)	~47 MB	95.7%	84.6%	64.8%
MobileFaceNet, 192-d (alternative)	—	98.3%	95.6%	88.9%
w600k_mbf, ArcFace MobileFaceNet (Phase 2)	**6.9 MB**	**99.4%**	**99.1%**	**98.0%**

Measured on the LFW benchmark. Duplicate-face detection at registration rose from about 65% to about 92% of same-person pairs. Field thresholds are tuned from on-device similarity logs.

Other engineering highlights

- **Memory on low-end phones:** fixed a TFLite native leak of ~60 MB per face scan, added a single-frame guard for ML Kit, and rewrote the inspection delete loader with SQL aggregates (10,000 records in ~0.5 s; before, 400 records took 6.6 s).
- **Enrollment quality:** Laplacian sharpness check and live focus feedback so blurry faces are never registered.
- **Beta (Phase 2 branch):** offline aerial map (PMTiles) with Strava-style route recording, findings and a ticket workflow; offline road routing (Dijkstra on GIS shapefile) with an OSRM online fallback; on-device PPE detection (YOLO, 6 classes, mAP50 0.83).

Tech: Flutter, Dart, GetX, Drift / SQLite, Dio, TensorFlow Lite, Google ML Kit, NFC (nfc_manager), Bluetooth thermal printer, Geolocator, Git

2. ePMS and eTis — Plantation & External Harvest Apps

PT Austindo Nusantara Jaya Tbk (ANJ Group) Dec 2024 – Aug 2025

- **ePMS:** planning, harvest and inspection for several estates, with end-to-end inspection assignments until each task is closed.
- **eTis (E-Trace):** lets plasma and external growers create harvest order tickets to sell palm fruit to ANJ. Released on the Play Store.
- Refactored data sync and local database fetching to cut sync time; maintained and released 2 apps.

Tech: Flutter, NFC, Firebase Cloud Messaging, AWS CodeCommit, Postman

3. ePCS — Electronic Plantation Control System

PT Lentera Kreasi Teknologi Sep 2025 – Nov 2025

Native Android app integrated with the PIMS backend. Reads TPH (harvest collection point) data from NFC tags to create harvest tickets and daily work realization.

Tech: Java, Android Studio, NFC, GitHub

4. WasteApp — Waste Bank App with Object Detection

Final Capstone, Telkom University · Team of mobile, backend and ML roles Nov 2023 – Aug 2024

Flutter app for a waste bank in Bandung that recognises bottles, plastic and other waste from a photo through a YOLOv8 REST API, then handles deposits and balances. Firebase Auth, Firestore and Storage; backend on Google Cloud. Published in IEEE and registered as intellectual property (HKI).

Tech: Flutter, Dart, Firebase, REST API, Figma, GCP · GitHub · Team repos · Play Store

5. Other Projects

- **FlixID (2024):** movie ticket app with the TMDB API, Firebase and Riverpod in clean architecture. GitHub
- **My-WB (2023):** Flutter waste detector with reuse, reduce and recycle video recommendations. GitHub
- **Cycle-Me (2023):** Bangkit capstone backend in Node.js / Express with a Flask ML service, deployed on App Engine, Cloud Run and Compute Engine. GitHub
- **Posyandu Alamanda (2022):** community health post website with mother and child service registration. Site